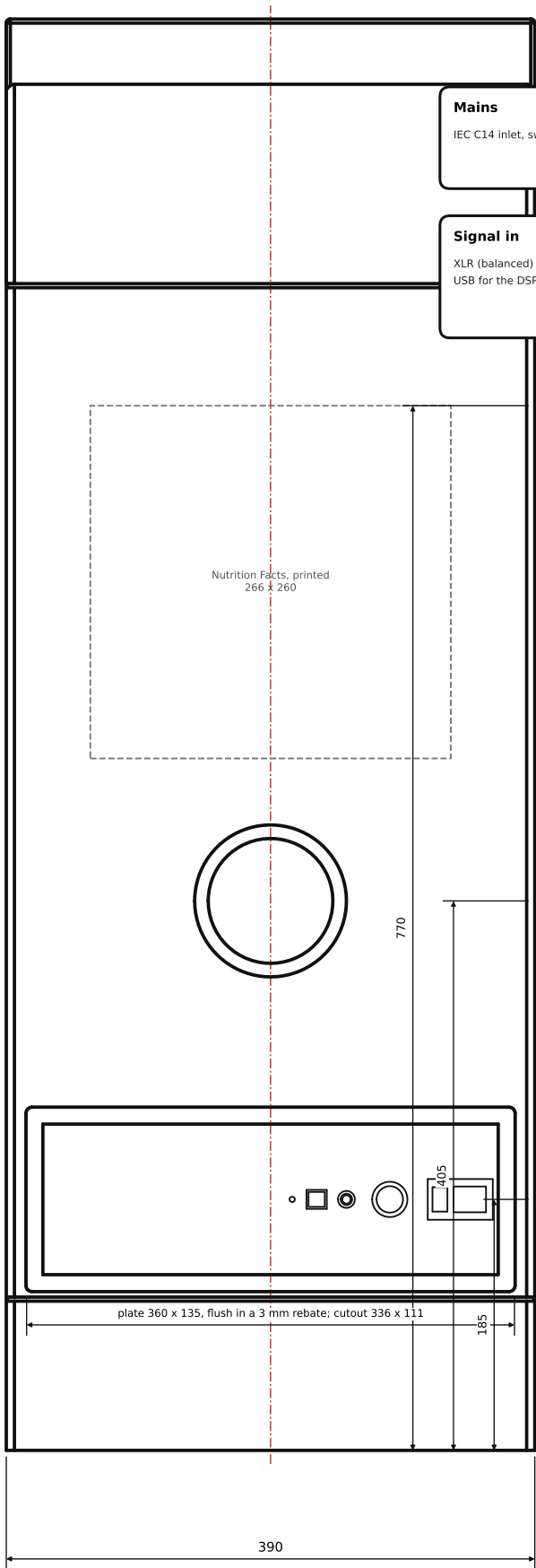


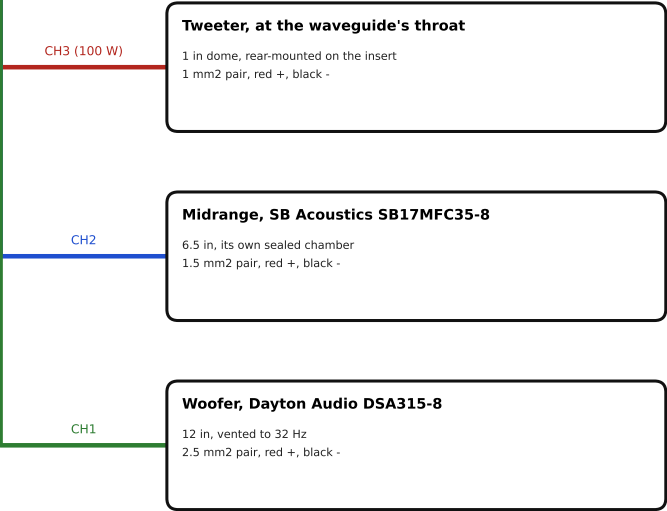


BACK

Mains
IEC C14 inlet, switch

Signal in
XLR (balanced) or RCA
USB for the DSP

Hypex FusionAmp FA253
3 channels, DSP crossover and EQ (ADAU1452)
250 + 250 W into 4 ohm, 100 W tweeter channel
on its side across the back's foot
in its own sealed box (not airtight itself)
filters: Hypex Filter Design, over USB
mains in on the plate (IEC C14 with switch)



Runs, connectors and seals

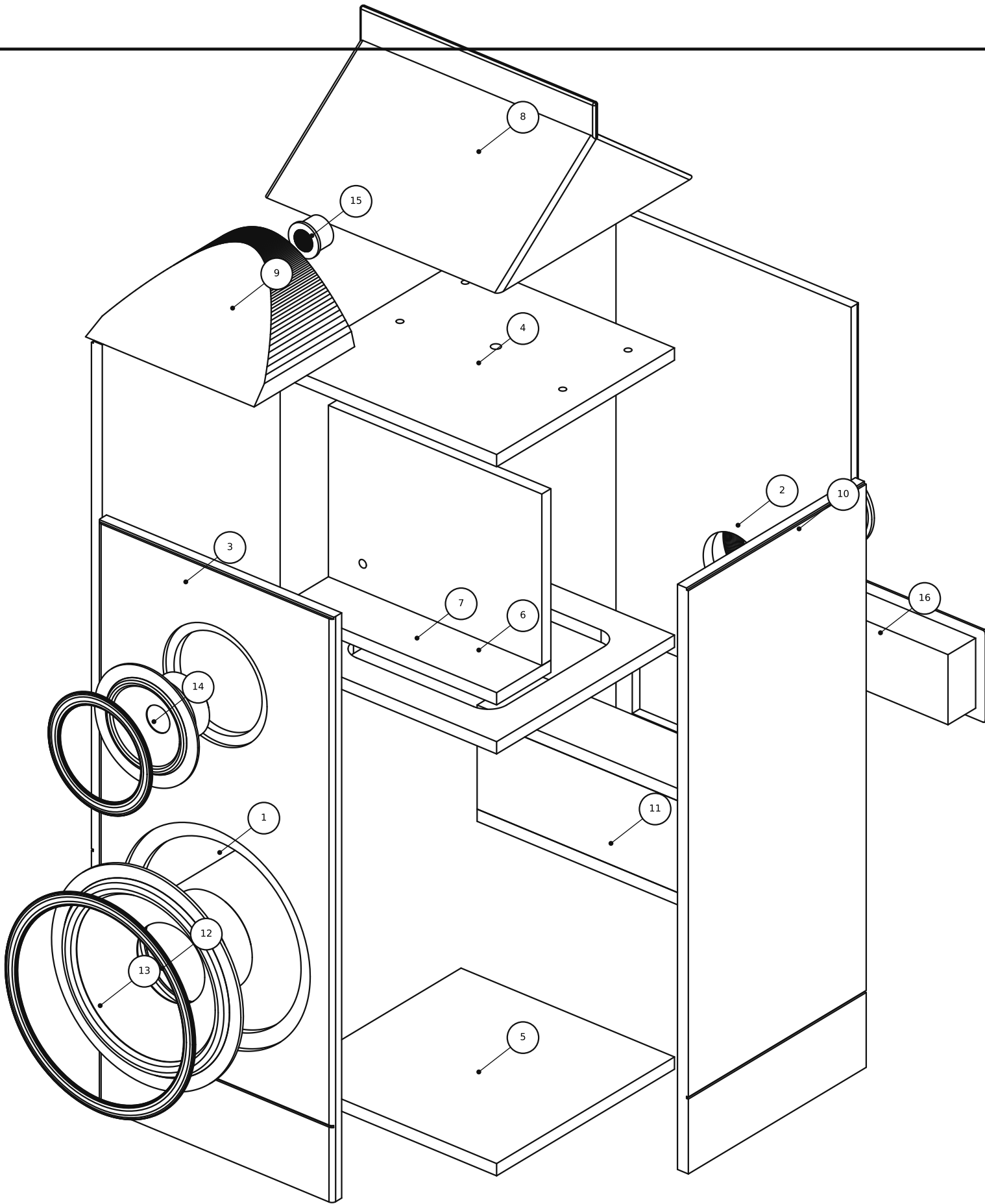
1. Woofer (CH1): from the amplifier box's gland straight to the woofer's terminals, 0.6 m of 2.5 mm2. Fit 6.3 mm push-on terminals to suit the driver.
2. Mid (CH2): up the back wall, through the 12 mm hole in the mid chamber's divider, 1.0 m of 1.5 mm2. Seal the hole round the cable with putty: the mid's chamber must stay closed.
3. Tweeter (CH3): up the back wall to the top panel, through the 14 mm channel in the top panel and the gable block, 1.3 m of 1.0 mm2, to a two-pole locking connector (JST VH or Molex Mini-Fit) at the insert. The supplied 1.25 m Hypex harness is too short: extend it.
4. Seal the tweeter cable where it passes the top panel with a rubber grommet and sealant. The insert's pocket is open to the room through its 0.3 mm seam, so an unsealed channel would be a leak in the woofer's box.
5. The amplifier box's lid: a 20 mm gland (PG11) for the three pairs; seal it. The box itself is glued and sealed; its front is the woofer chamber's wall.
6. Polarity, delay and the crossover (about 2.8 kHz tweeter to mid, from the waveguide study) are set in the DSP: wire red to + throughout and let the filters do the rest.

earmilk floorstander | Back and wiring

The back at 1:5 and the amplifier, drivers, cables and connectors. Active: the amplifier's DSP is the crossover.

mm | first issue 2026-10-08 | from fab/sheets.py (the CAD)

Check every bought part against its own drawing before cutting.



- Parts**
- 1 **Front baffle x1**
18 mm Baltic birch, CNC: cutouts and rebates for the drivers
 - 2 **Back x1**
18 mm birch: port hole, the amplifier's cutout and rebate
 - 3 **Sides x2**
18 mm birch
 - 4 **Top panel x1**
18 mm birch, the tweeter cable's 14 mm hole, four dowel holes
 - 5 **Bottom panel x1**
18 mm birch
 - 6 **Window brace x1**
18 mm birch
 - 7 **Mid chamber shelf and divider x2**
18 mm birch; the divider's 12 mm cable hole
 - 8 **Gable block x1**
laminated birch, CNC: the insert's pocket and the cable channel
 - 9 **Waveguide insert x1**
CNC from solid (MDF, HDPE) or printed (SLA, MJF); magnets, pins
 - 10 **Port x1**
printed tube and flare collar, 92 bore
 - 11 **Amplifier box x3**
18 mm birch floor, lid (gland) and front, glued and sealed
 - 12 **Woofer x1**
Dayton Audio DSA315-8, 12 in (PLACEHOLDER until confirmed)
 - 13 **Trim rings x2**
printed, satin black or gunmetal, over each frame and its screws
 - 14 **Midrange x1**
SB Acoustics SB17MFC35-8, 6.5 in
 - 15 **Tweeter x1**
1 in dome, rear-mounted on the insert (choice pending)
 - 16 **Amplifier x1**
Hypex FusionAmp FA253, DSP, 3 channels

earmilk floorstander | Exploded view and parts

The parts pulled apart along the way they go together, and what each one is.

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